CS2201 Computer Organization & Architecture

L32: Introduction to Network on Chip



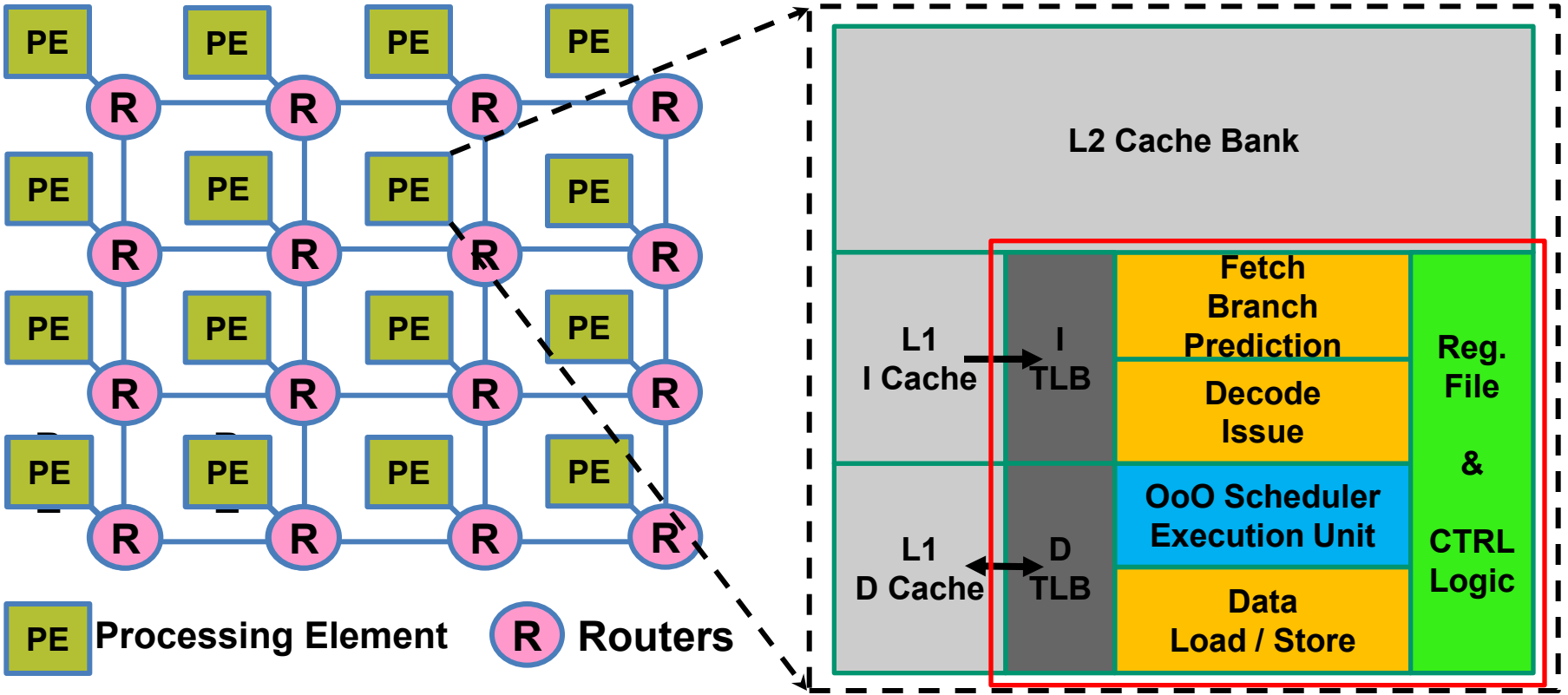
John Jose

Professor

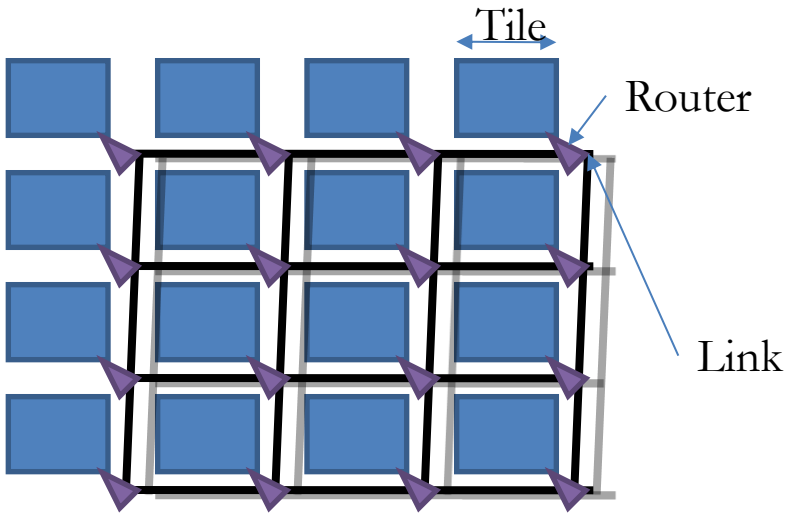
Department of Computer Science & Engineering

Indian Institute of Technology Guwahati

Tiled Chip Many-Core Processor (TCMP)



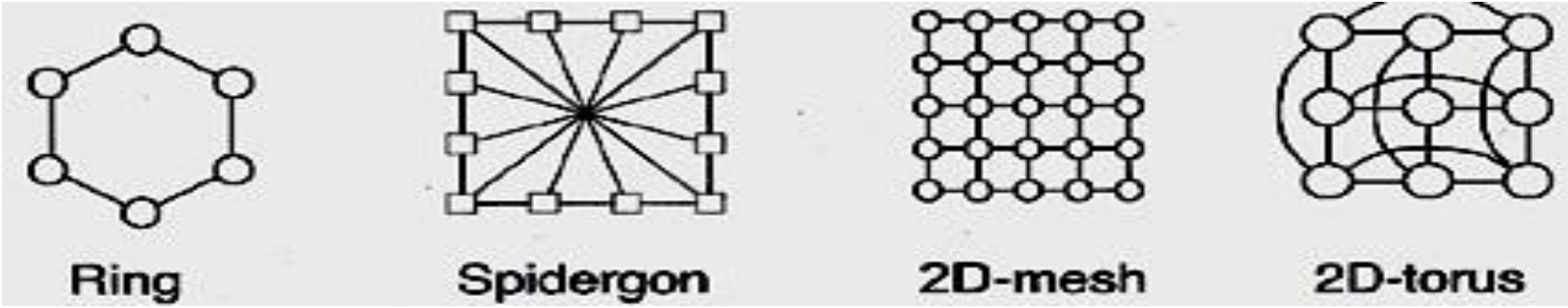
Building Blocks of NoC



- Topology
- Routing
- Flow control
- Router micro-architecture

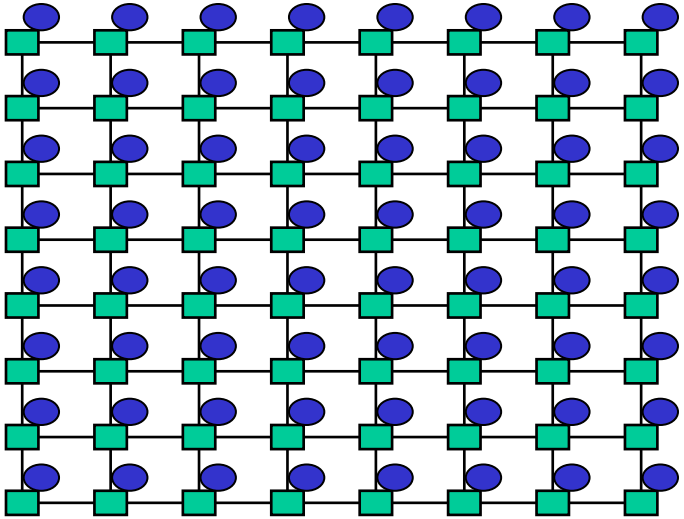
Topology

- Determines the physical layout and connection pattern between nodes and channels in the network.

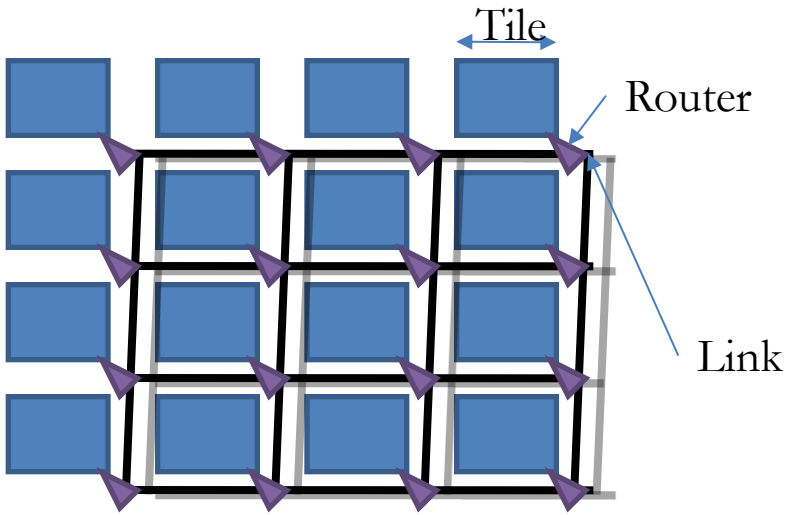


Mesh

- Each node connected to 4 neighbors (N, E, S, W)
- Easy to layout on-chip: regular and equal-length links
- Path diversity: many ways to get from one node to another



Building Blocks of NoC



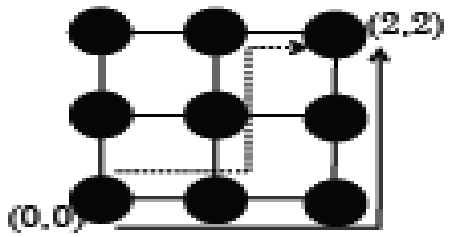
- Topology
- Routing
- Flow control
- Router micro-architecture

Routing Algorithm

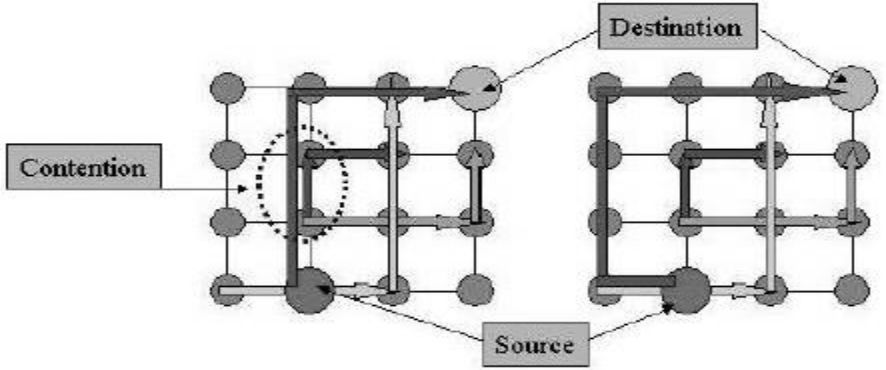
- Compute the path route for packets to reach destination.
- Deterministic Routing vs Adaptive Routing
  - **Deterministic:** always chooses the same path for a communicating source-destination pair
  - **Oblivious:** chooses different paths, without considering network state
  - **Adaptive:** can choose different paths, adapting to the state of the network

Minimal & Non-Minimal Routing

- **Profitable route:** The route that always leads the packet closer to the destination.
- **Misroute:** A route that leads the packet away from the destination.



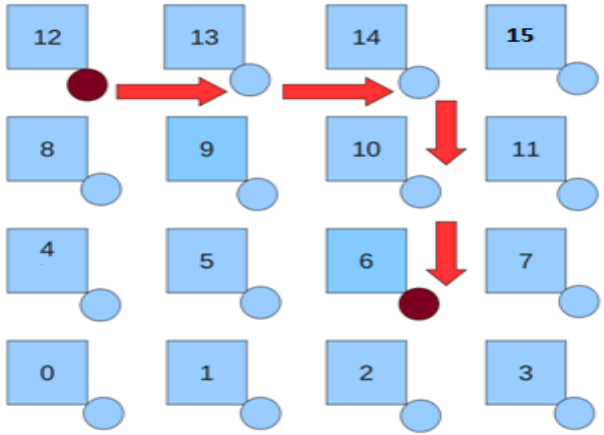
Minimal routing



Non-Minimal routing

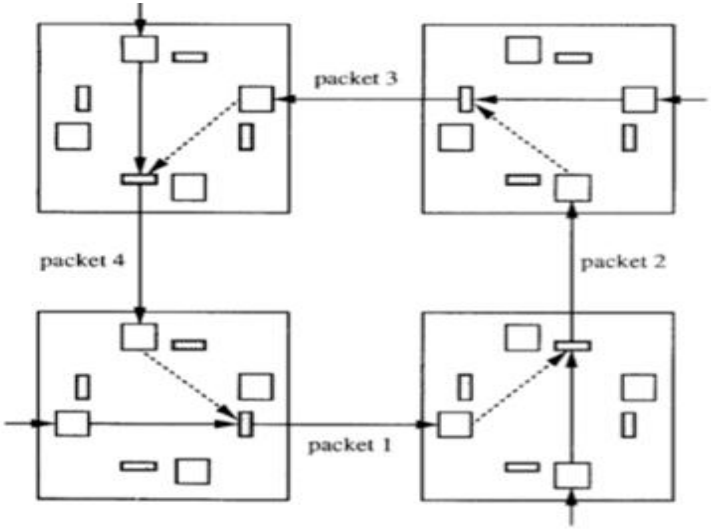
Deterministic Routing

- All packets between the same (source, dest) pair take the same path
- Dimension-order routing – Eg. XY routing
  - First traverse dimension X, then traverse dimension Y
  - Simple
  - Deadlock freedom
  - Could lead to high contention
  - Does not exploit path diversity



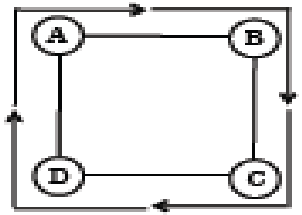
Deadlock

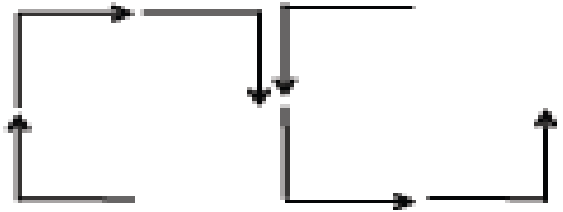
- No forward progress
- Caused by circular dependencies on resources
- Each packet waits for a buffer occupied by another packet downstream



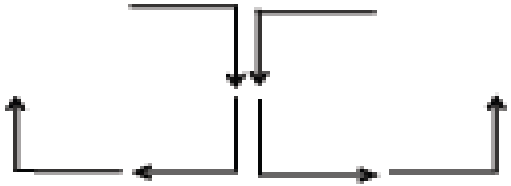
Turn Model to Avoid Deadlock

- Analyze directions in which packets can turn in the network
- Determine turns the can form cycles
- Prohibit just enough turns to break possible cycles

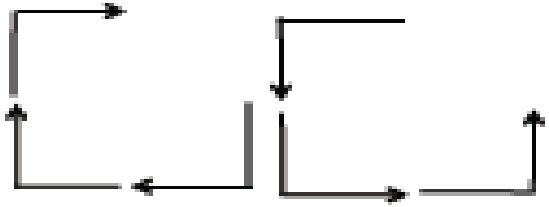




West First Turns

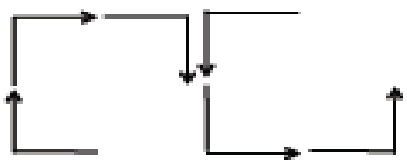


North Last Turns

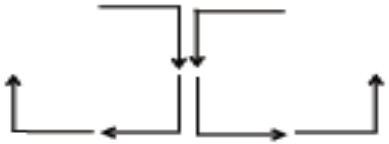


Negative First Turns

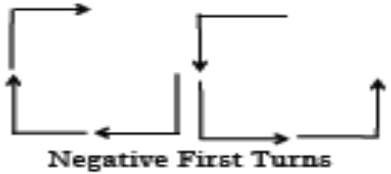
Turn Model to Avoid Deadlock



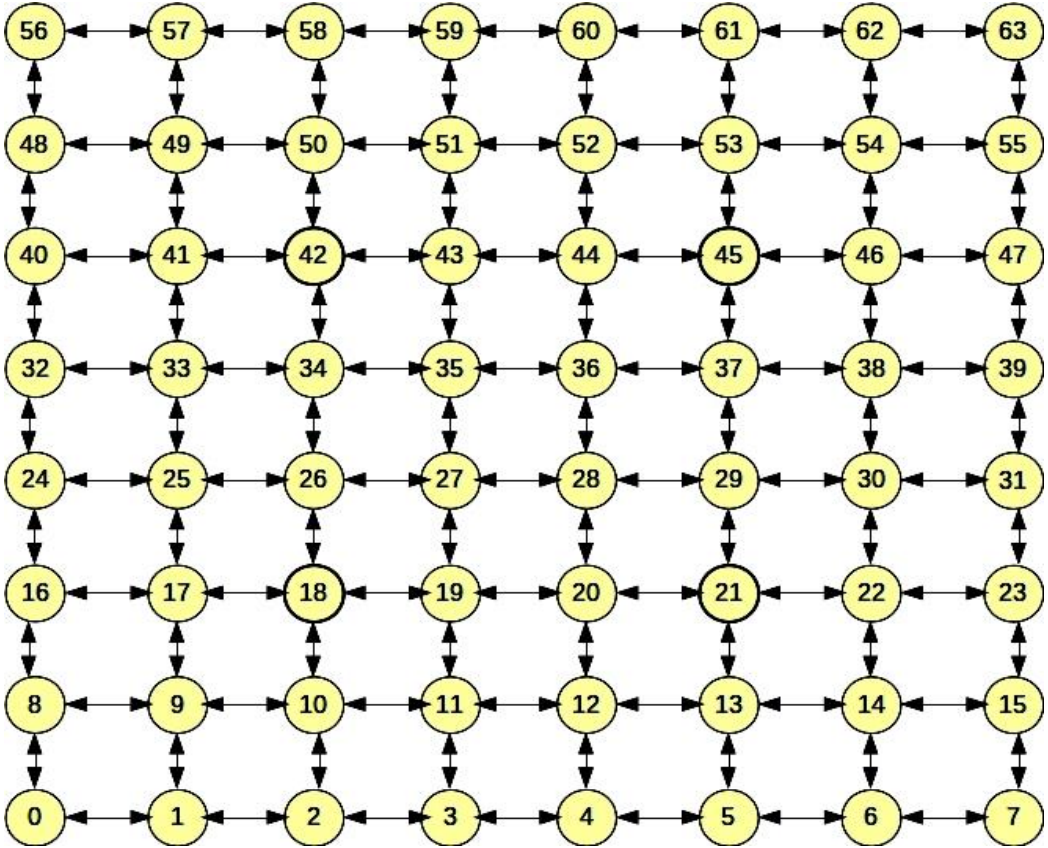
West First Turns



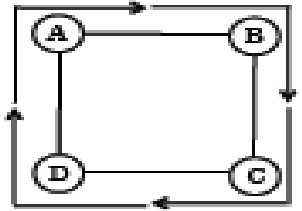
North Last Turns



Negative First Turns

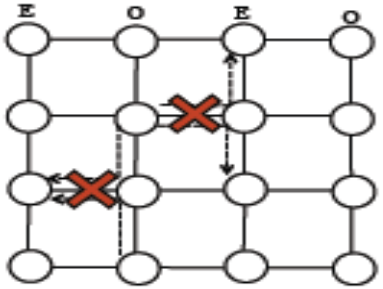


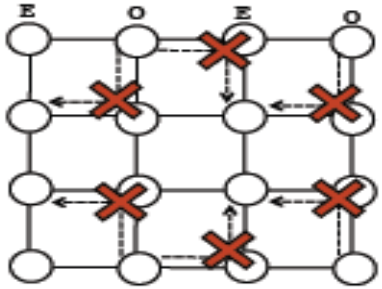
Adaptive Odd-Even Turn Routing

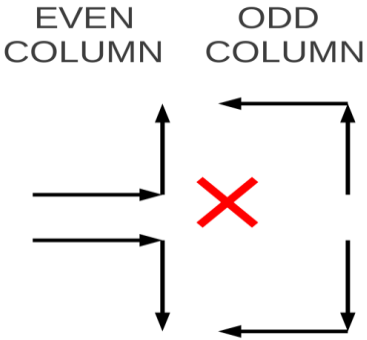


Non restrictive turns leads to deadlocks

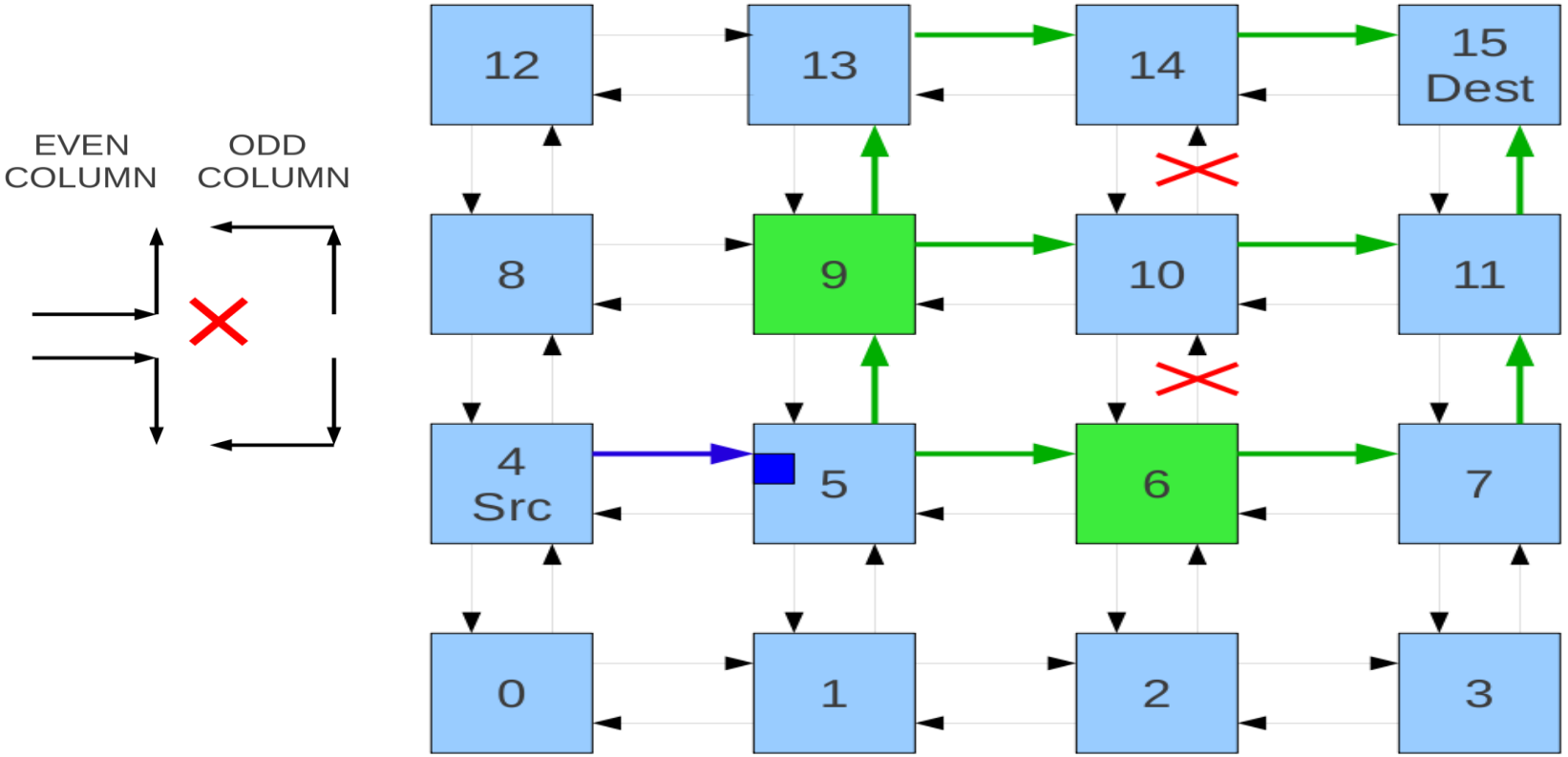
- Prohibited Turns at certain junctions
- For nodes in even column EN and ES.
- For nodes in odd column NW and SW.



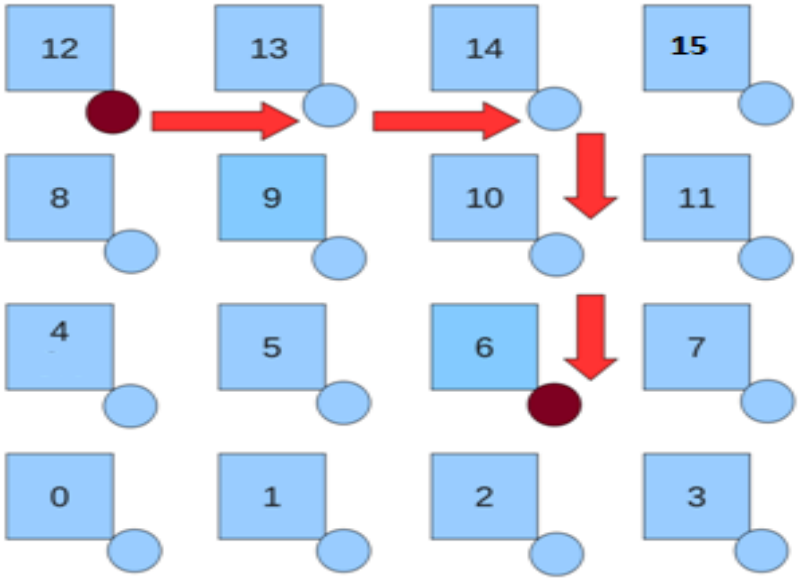




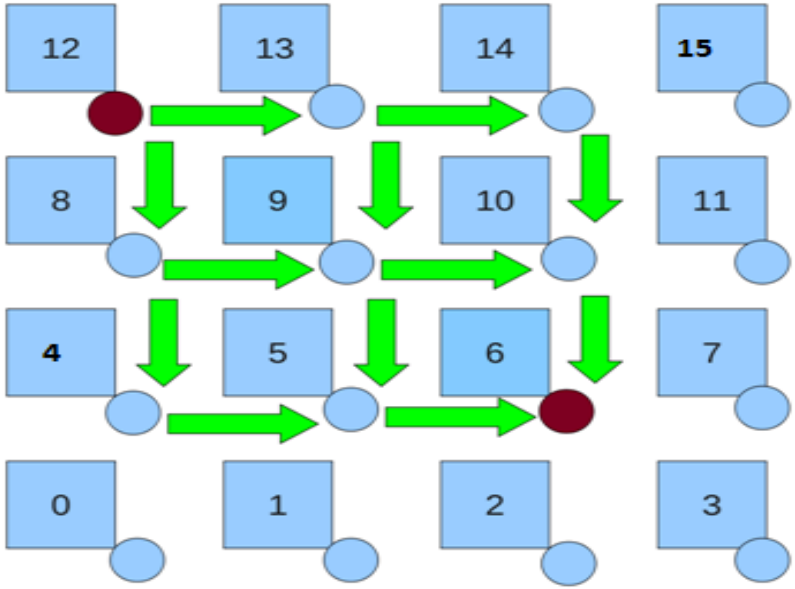
The Minimal Odd-Even Routing



Static vs Adaptive Routing



Static routing – XY routing

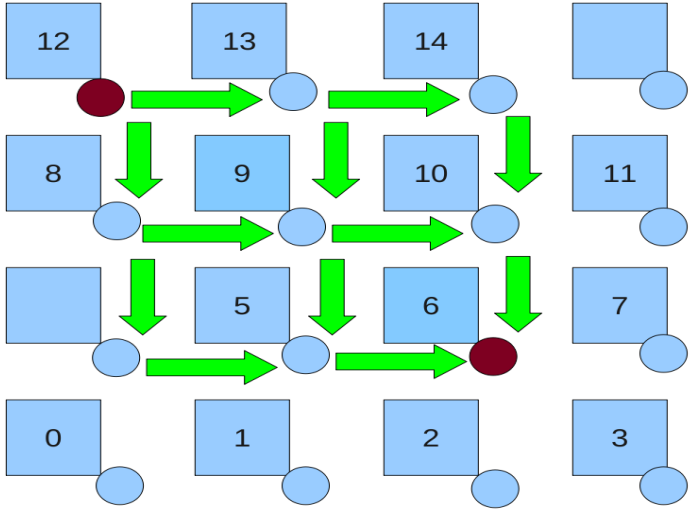


Adaptive Routing

Adaptive Routing

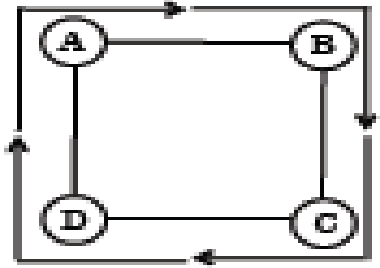
- Router uses network state (e.g., downstream buffer occupancy) to pick which productive output port to send
- Productive output port - port that gets the packet closer to its destination
- Aware of local congestion

Adaptive Routing

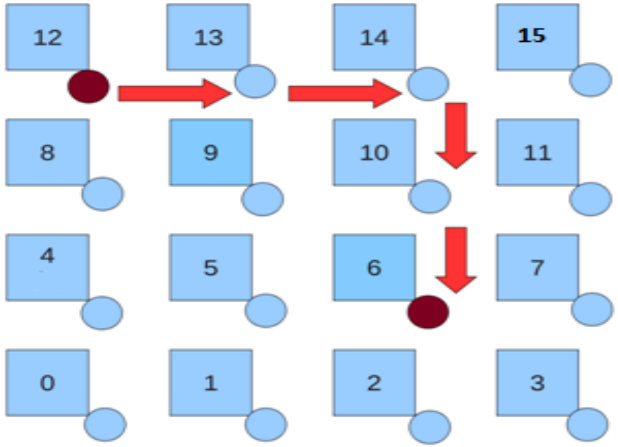


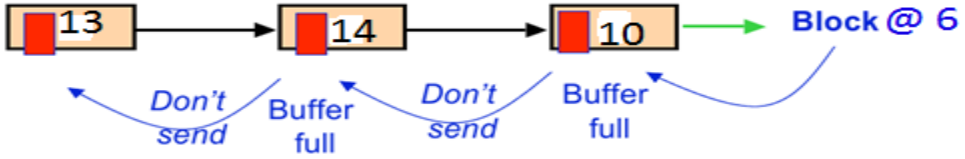
Minimal- Adaptive Routing

- Local and global information can be exploited for route selection.
- Preserving inter message order and out of order delivery.
- Should be deadlock and livelock free.

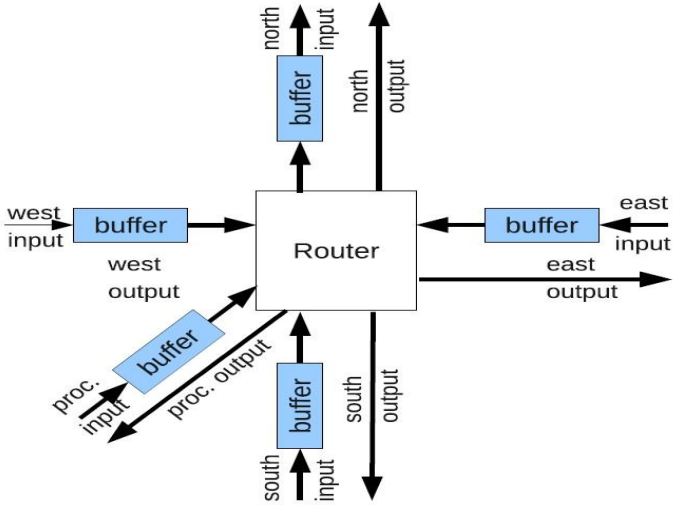


Flow Control



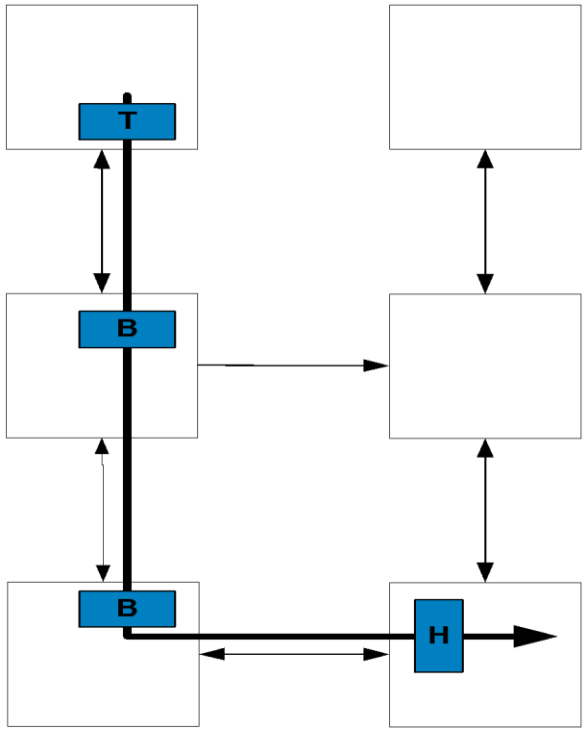


"Backpressure"



- Upstream router should know the buffer availability of downstream router.
- Credit should be exchanged between routers by handshake signals

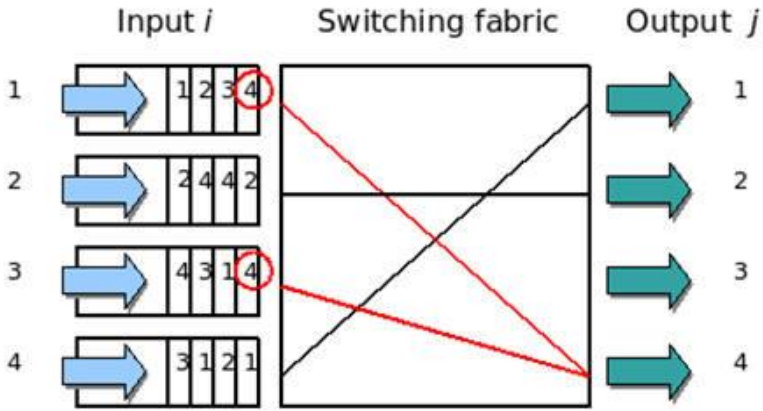
Wormhole Flow Control



- Packets broken into smaller flits
- Flits are sent across the fabric in a *wormhole fashion*
  - Body follows head, tail follows body
  - Pipelined
  - If head blocked, rest of packet stops
  - Routing (src/dest) information only in head
- Lower latency, efficient buffer utilization
- Occupies resources across multiple routers

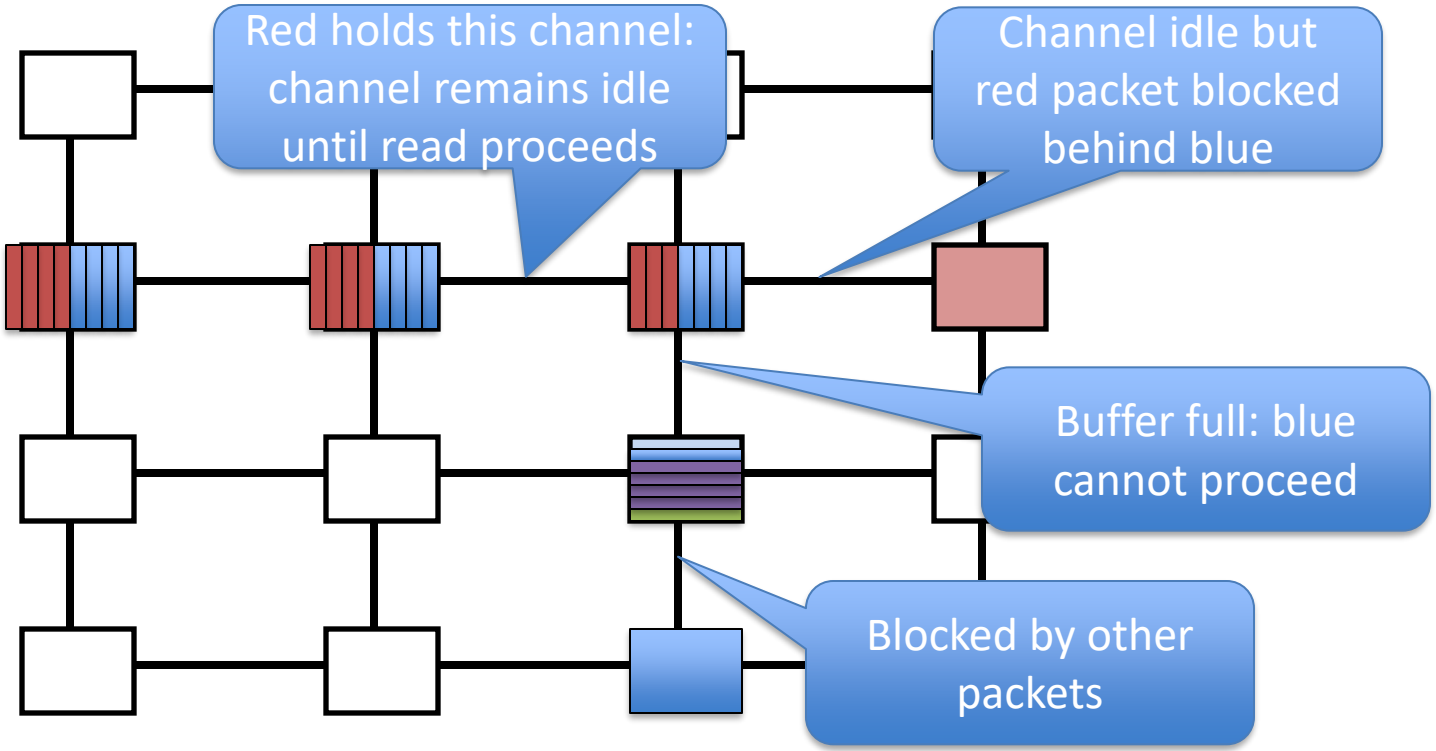
Head of Line Blocking

- Suffers from **head of line blocking**
  - If head flit cannot move due to contention, another worm cannot proceed even though links may be idle

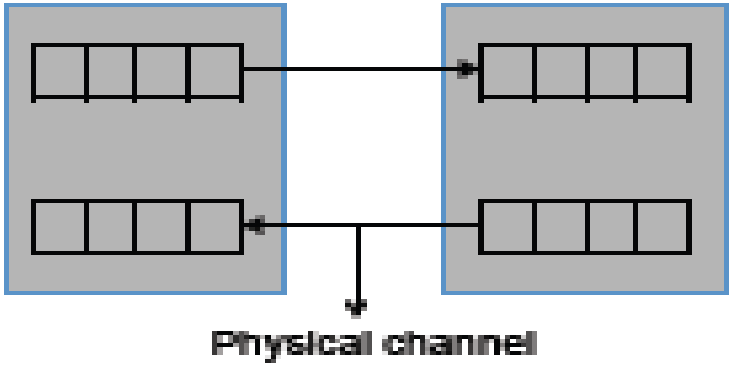


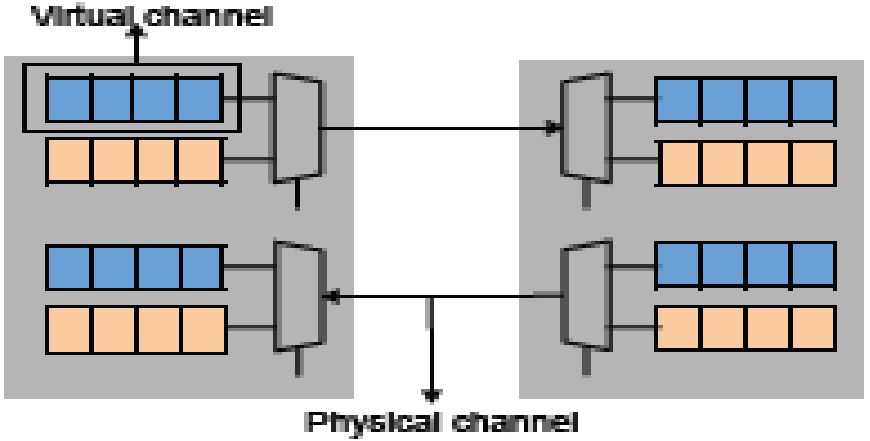
HOL Blocking

Head of Line Blocking



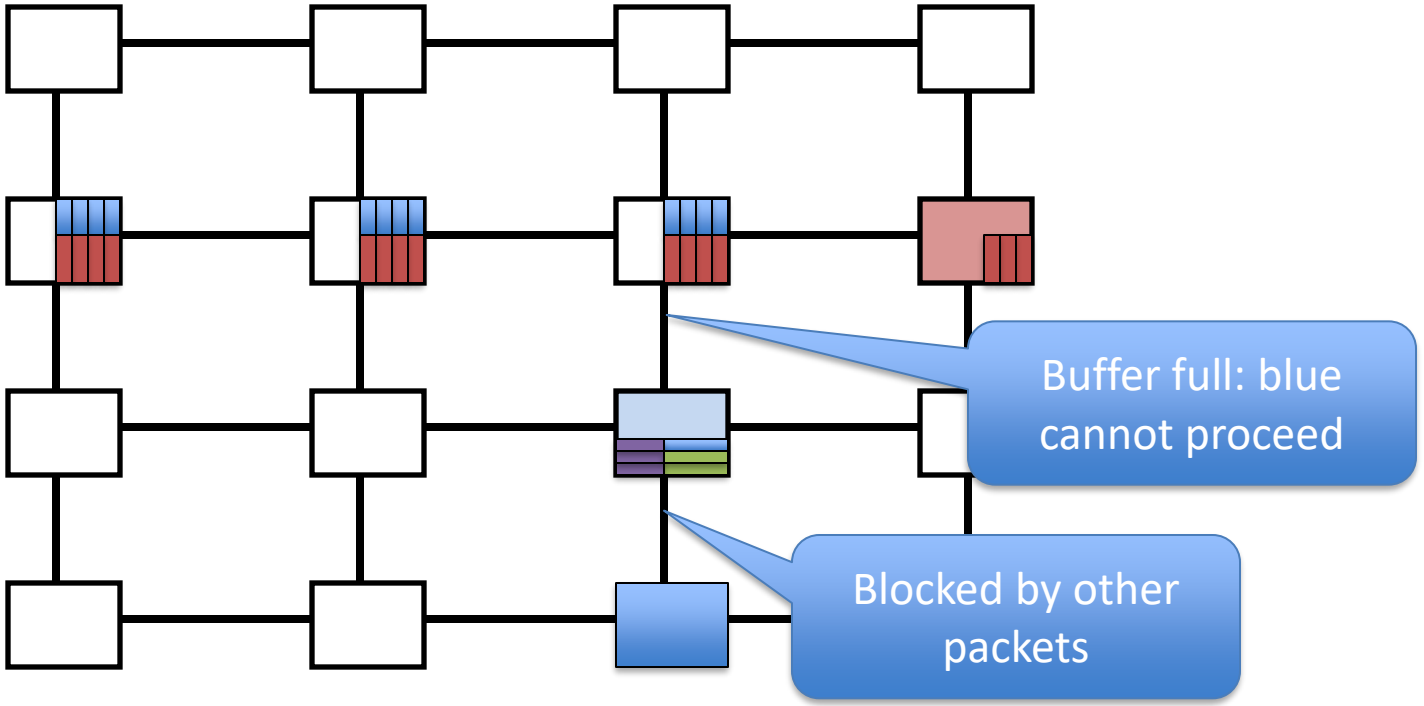
Virtual Channel Flow Control



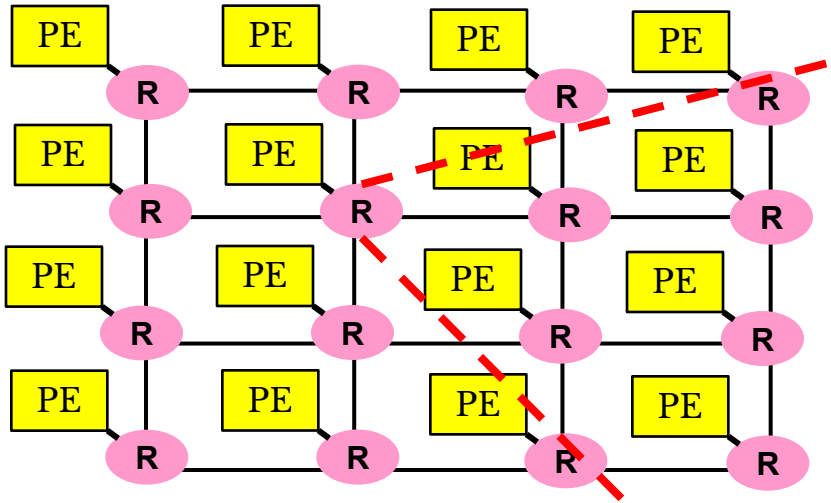


- VCs are allocated once at each router to the head flit and remaining flits of the packet inherit the same VC
- Flits of different packets can be interleaved on the same physical channel
- VCs avoid deadlocks

Virtual Channel Flow Control



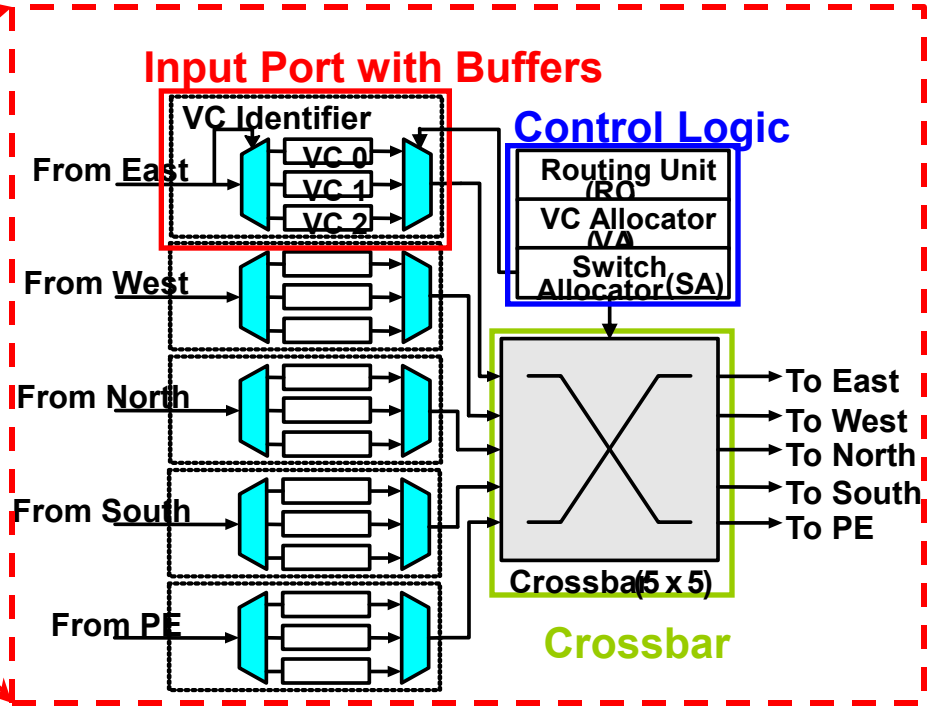
Input Buffered NoC Router



R Router

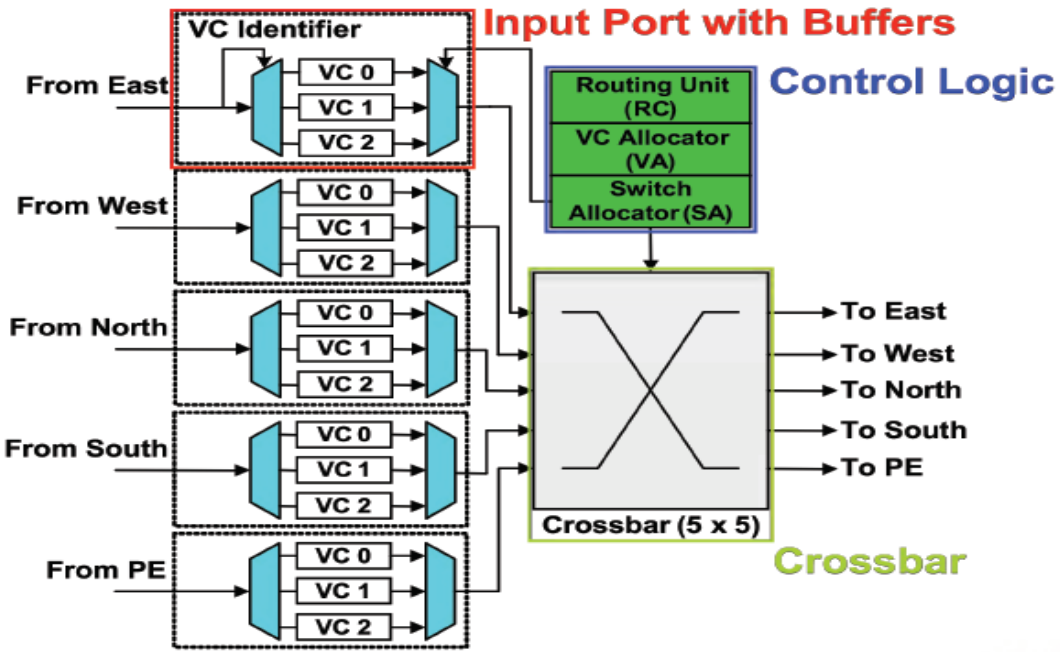
PE Processing Element

(Cores, L2 Banks, Memory Controllers etc)



Functions of a Router

- Buffering of flits
- Route computation
- VC allocation
- Switch Allocation
- Switch Traversal

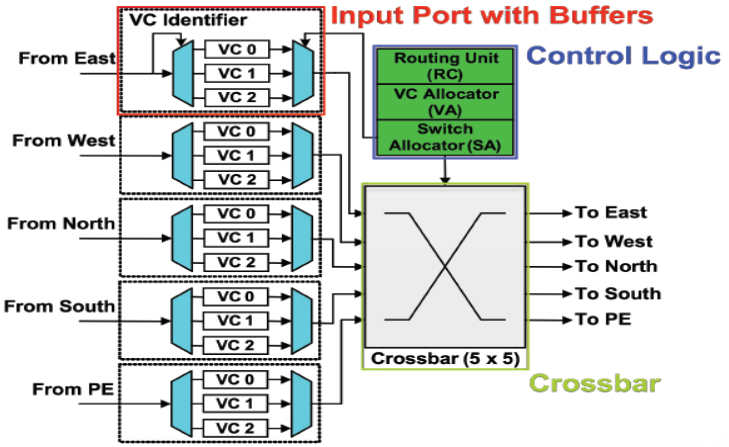


- Link Traversal

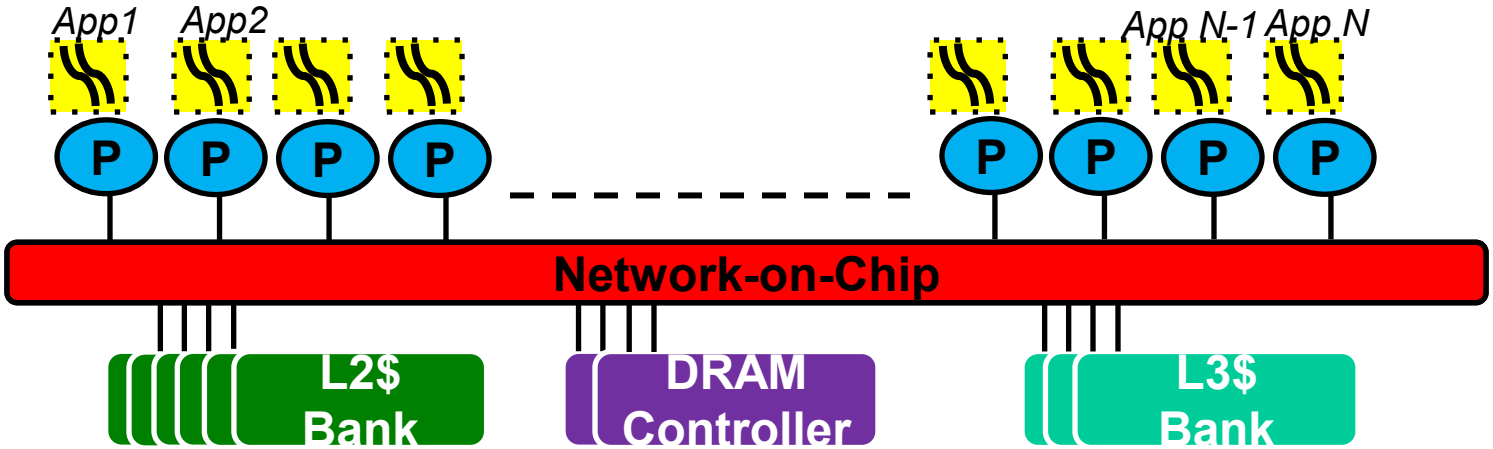
Router Pipeline



- Five logical stages
  - BW: Buffer Write
  - RC: Route computation
  - VA: Virtual Channel Allocation
  - SA: Switch Allocation
  - ST: Switch Traversal



How critical is NoC ?



Network-on-Chip is a critical resource shared by multiple applications



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